

PHARM:
Estrogens and Progestins

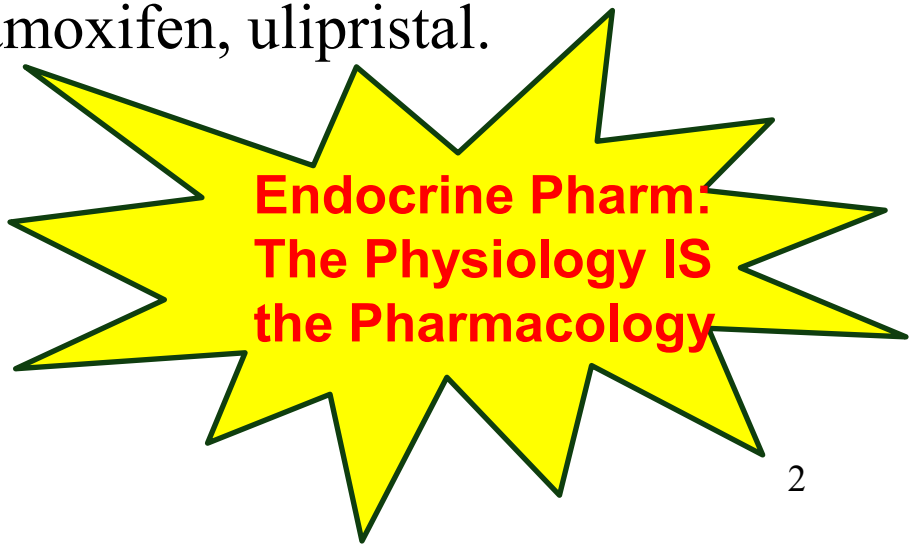
Carol L. Beck, Pharm.D., PhD

The lecturer has no conflicts of interest related to this topic.

Learning Objectives

By exam time, you should be able to:

1. Explain how the structure and function of the estrogen and progesterone receptors affect drug actions and tissue specificity
2. Describe the mechanism of action, signature adverse effects, and contraindications of: fulvestrant, mifepristone; raloxifene, tamoxifen, ulipristal.



Overview

Estrogens Any of various natural steroids (such as estradiol) that are formed from androgen precursors, that are secreted mainly by the ovaries, placenta, adipose tissue, and testes.

- Estrogen is one of two dominant sex hormones in people with functioning ovaries.

- In people with ovaries, it stimulates the development of secondary sex characteristics and promotes the growth and maintenance of the reproductive system
- In people with testes, it modulates libido, erectile function, and spermatogenesis

Overview

Progestogens

"Promote gestation"

- A steroid sex hormone $C_{21}H_{30}O_2$ that is secreted by the corpus luteum to prepare the endometrium for implantation and later by the placenta during pregnancy to prevent rejection of the developing embryo or fetus.
- Also a precursor to testosterone

Progestins --synthetic

Progesterone -- natural

I. Estrogen and Progesterone

Remember:

Estrogen and ANDROGEN (testosterone) are the Sex Steroids

- Bound by Sex Steroid Binding Globulin (SSBG)

Progesterone and Cortisol

- Bound by Transcortin aka Corticosteroid Binding Globulin

Only free drug/ hormone available to bind to receptors and produce response

I. Estrogen and Progesterone

A. Steroid structure

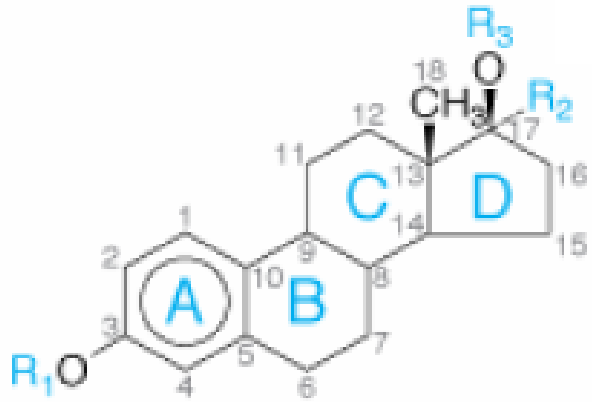
- Lipids with three 6-C rings and one 5-C ring
- Five classes of steroid hormones
  - Estrogens
  - Progestogens
  - Androgens
  - Mineralocorticoids
  - Glucocorticoids

ProgestOgen or ProgestAgen or Progestin
You will see it spelled all of these ways.

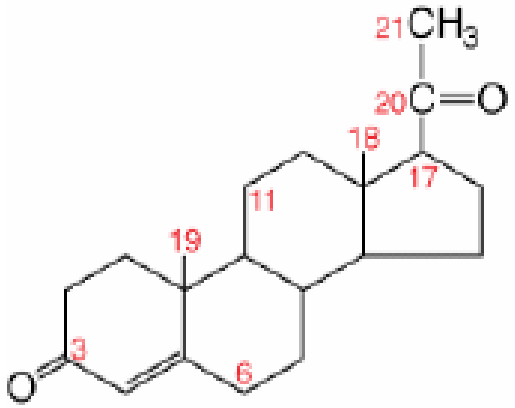
You will have lectures on Steroidogenesis

Structure

STEROIDAL* ESTROGEN



PROGESTERONE



*Yes, there are non-steroidal estrogens

Progestin
+
Luteo-*sterone*

→

Combined Name

Proge*sterone*

B. Metabolism/ Inactivation

- Inactivated by conjugation to sulfate esters or glucuronic acids
- Metabolized by oxidation– primarily in liver
- Excreted in urine

C. Biosynthesis

Background

1. Progesterone (natural progestin)

Made in: ovaries, placenta, adrenal gland, testis

- Ovaries (primarily corpus luteum) and placenta make cholesterol → progesterone

- Adrenal gland makes cholesterol --> progesterone -- testosterone

- Testis makes cholesterol --> progesterone --> testosterone

C. Biosynthesis

Background

2. Estrogen

Made in: ovaries, peripheral tissue, liver

- Ovaries* (first in developing follicles then in corpus luteum)
- Peripheral tissue (primarily adipose tissue) makes androstenedione which is converted to estradiol --> estriol and estrone
- Liver makes estradiol which is converted to estriol and estrone

*Primary source of sex hormones between puberty and menopause.

2. Estrogen

Protein binding:

95% of estrogens circulate attached to proteins, primarily sex hormone binding globulin (SHBG) and albumin.

- ~5% is free/unbound

- SHBG has low capacity for binding, but high affinity; albumin has high capacity but low affinity.

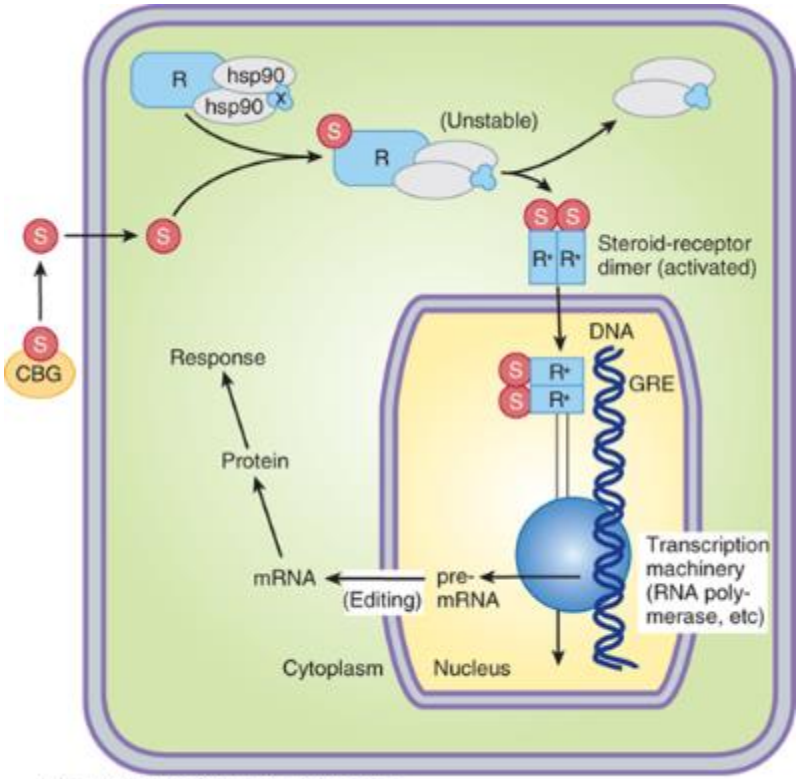
- Decreased SHBG leads to increased unbound circulating estrogens and androgens.

3. Neuroendocrine Regulation of Steroid Production

a. Hypothalamic secretion of **GnRH** acts on pituitary to release gonadotropins (**FSH** and **LH**)

b. Anterior Pituitary produces **FSH** (follicle stimulating hormone) and **LH** (luteinizing hormone) which act on the ovary and testis to stimulate steroid synthesis and gamete development.
   - FSH and LH on Ovary: Estrogen, Progesterone
   - LH on Testis: Testosterone (androgen)

D. Pharmacodynamics
Molecular Mechanism of Steroid Action



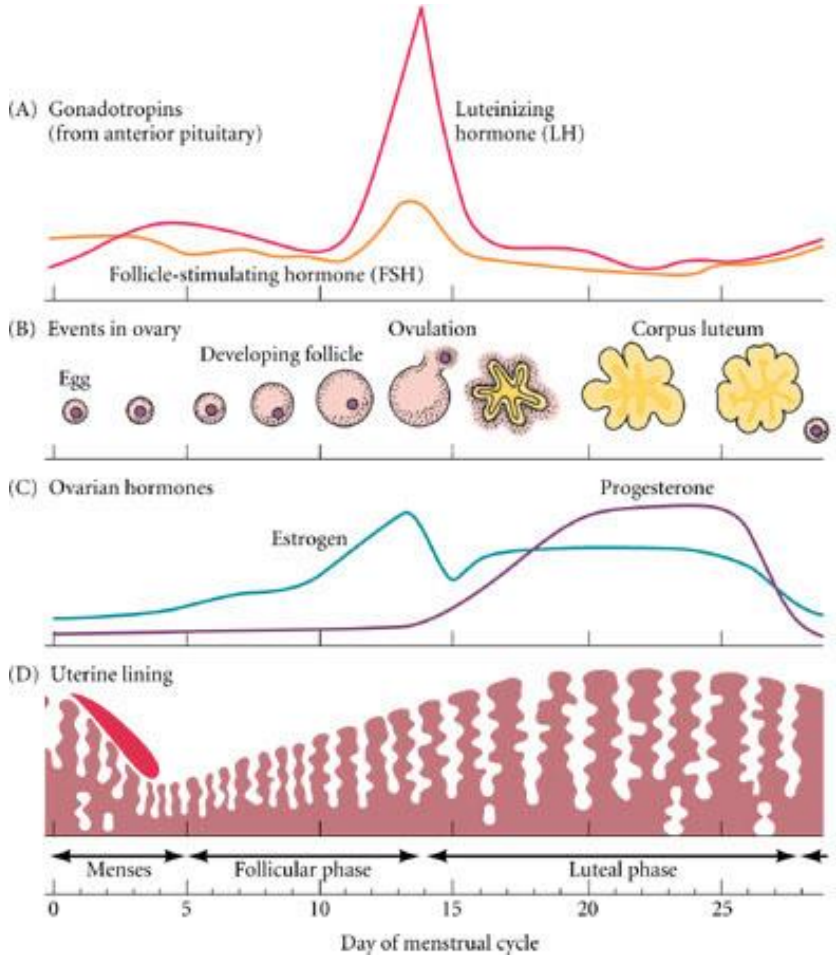
Source: Bertram G. Katzung, Todd W. Vanderah:
Basic & Clinical Pharmacology, Fifteenth Edition
Copyright © McGraw-Hill Education. All rights reserved.

a. Steroid hormones dissociate from binding protein and enter most cells by diffusion
b. In target cells, the steroid binds to receptors located in the cytoplasm and/or the nucleus (AKA “nuclear receptors”)
c. Steroid receptor is activated by hormone (ligand); dimerizes; binds DNA; induces transcription.

More later about Estrogen and Progesterone receptors!

E. Reproductive system

Review



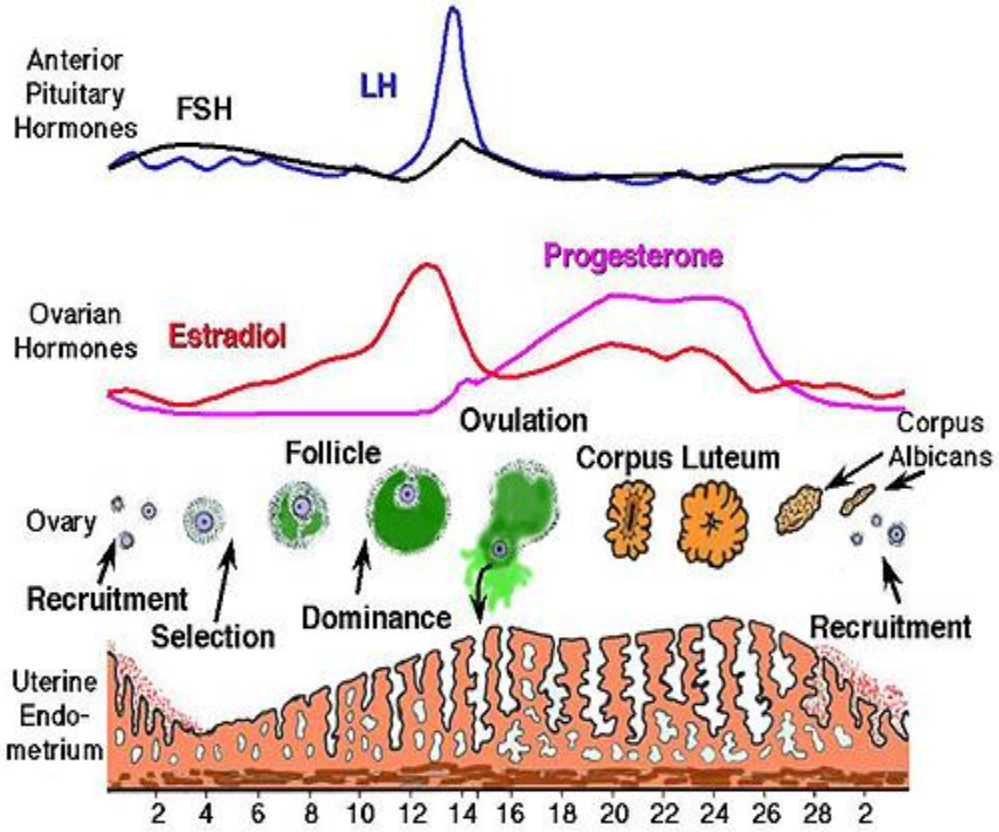
The menstrual cycle depends on a complex interaction between hypothalamus, pituitary, ovaries, and endometrium.

F. Systemic Effects of Estrogen
Natal Female

1. Anabolic
   - Development of vagina, uterus and tubes
   - Development of secondary sex characteristics
   - Pubertal growth spurt
2. Reproductive system
   - Regulation of menstrual cycle
   - Proliferation of uterine and vaginal epithelium
   - Stimulates production of cervical mucus, making it less viscous

Review

Physiologic Effects of Estrogen – Menstrual Cycle



For birth control: continuous estrogen– suppresses spike, no LH spike, no ovulation

F. Systemic Effects of Estrogen
Natal Male

1. Stimulates growth of stromal cells in prostate
2. Essential for spermatogenesis

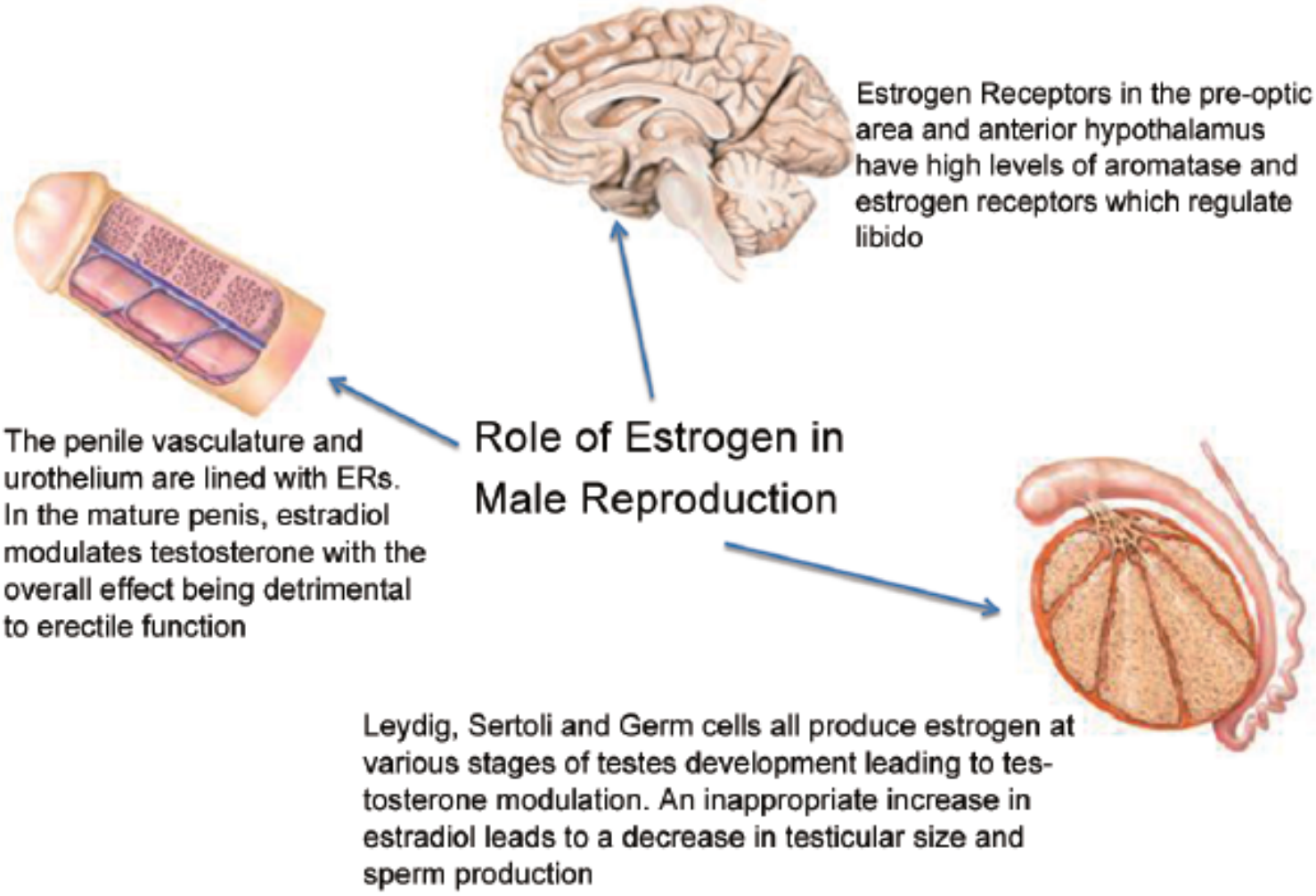


Figure 1: The role of estrogen in male reproduction.

Asian Journal of Andrology (2016) 18: 435-440

F. Systemic Effects of Estrogen

General

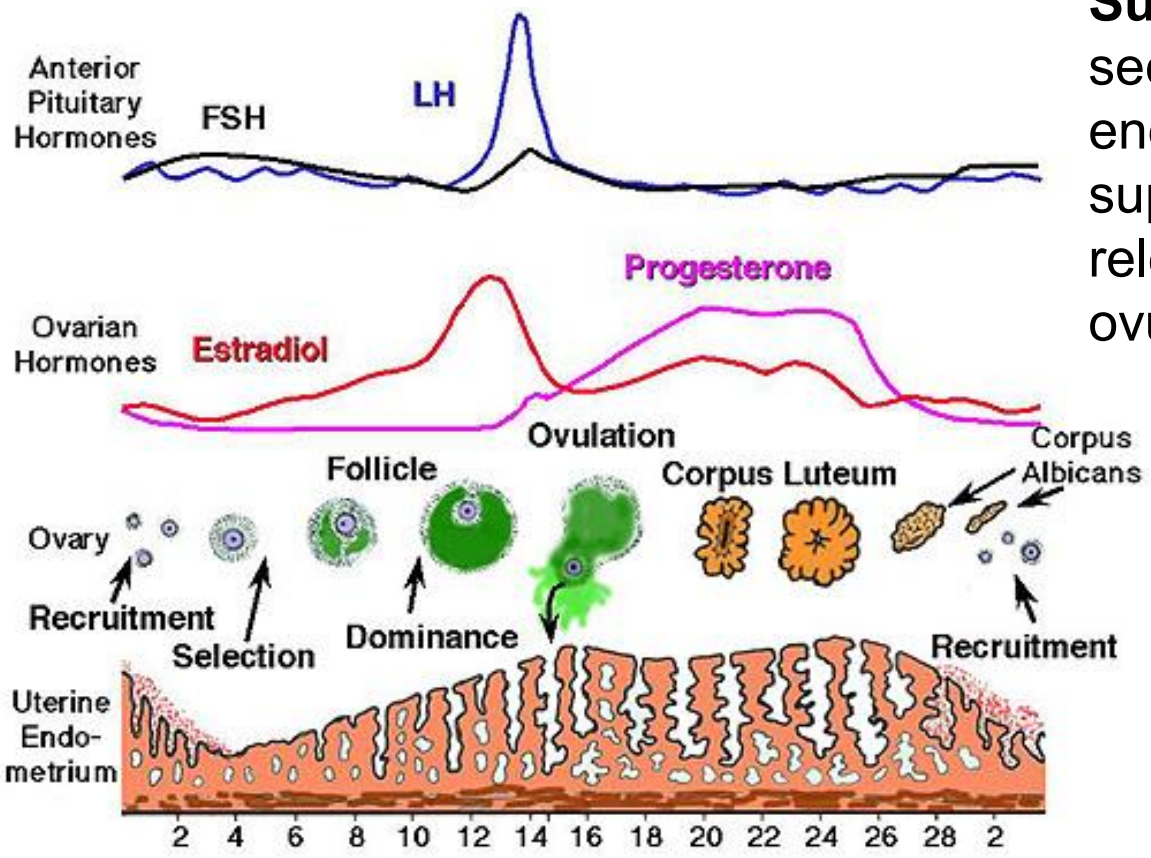
1. Blood Chemistry: effects on lipids
2. Suppresses cellular immune responses
3. Increases coagulation factors (increased coagulability)
4. Bone: Decreases rate of bone resorption by decreasing osteoclast activity and number

G. Systemic Effects of Progesterone
Natal Female

1. Reproductive system
   - Regulation of menstrual cycle
   - Modulates secretory differentiation and increases vascularity in uterus
   - Maintains pregnancy
   - Represses uterine contractions (smooth muscle relaxant)

Physiologic Effects of Progesterone – Menstrual Cycle

Review



Summary: Induces secretory change in the endometrium. At high doses suppresses gonadotropin release and suppresses ovulation

For birth control: continuous progesterone– no ovulation… and changes to endometrium do not occur

G. Systemic Effects of Progesterone

General

1. Blood chemistry: effects on lipids (HDL, LDL)

2. Endocrine: insulin levels and glucose response

3. Other: Body temperature; CO_2 response

II. Estrogen and Progesterone Receptors

Both are nuclear receptors!!!

A. Estrogen Receptors

1. **Human estrogen receptors: alpha and beta**

- Different genes on different chromosomes
- Different tissue/cell distribution
- Different affinity for ligands
- Different genes activated

A. Estrogen Receptors

2. **Estrogen Receptor activity**

- ER in many body tissues
- ER-Alpha and ER-beta levels depend on tissue
  - ER-Alpha: reproductive system
  - ER-beta: bone, brain, urinary tract

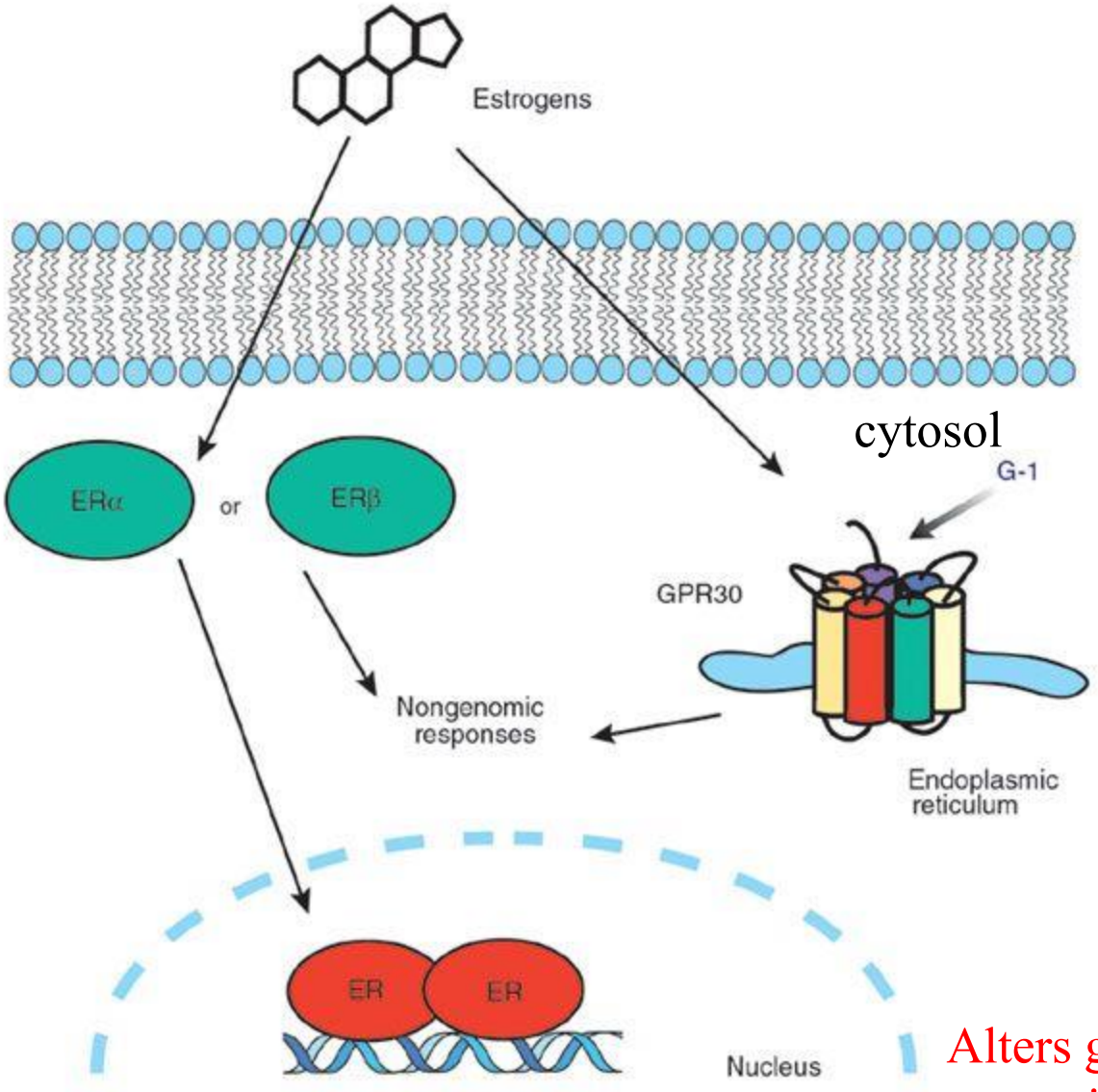
A. Estrogen Receptors

3. **Ligands for estrogen receptors**

- Estrogens: estradiol
  - binds to both alpha and beta
- Designed to be selective
- Selective Estrogen Receptor Modulators aka

  SERMS: tamoxifen, faloxifene, toremifene
- Novel ER-alpha and ER-beta selective ligands

Ligand



Receptors

GPR30 is a G protein coupled receptor that binds estrogens and antiestrogens; may explain some of the faster responses to estrogens

Receptor Dimer

Binds to hormone response elements in DNA;

Alters gene transcription

Weatherman RV. Nature Chem Bio, 2: 175-6, 2006

B. Progesterone Receptors

1. **Progesterone receptors**: alpha and beta, also called PRA and PRB
   - PR beta has additional 164 N-terminal aa
   - Different transcriptional activities
   - Cell type and promoter specific effects– often PR-beta > PR alpha
   - PR alpha can repress PR beta and ER signaling
   - (Less known about PR compared to ER)

B. Progesterone Receptors

2. **Ligands** for Progesterone receptors

- Progesterones: progestins (example: norethindrone)
- Selective Progesterone Receptor Modulators aka

  SPRMS: ulipristal

C. ER/PR Pharmacology/ Pharmacodynamics

- In uterus PR opposes ER stimulation
- In breast PR increases ER stimulation

C. ER/PR Pharmacology/ Pharmacodynamics

Ligands can be:

- Agonists
- SERMs/SPRMs (mixed agonist/antagonist)
  - Selective Estrogen Receptor Modulator
  - Selective Progesterone Receptor Modulator
- Antagonists

ER/PR Pharmacology/ Pharmacodynamics

- Different ligands induce different receptor confirmations that determine response
- Partner proteins/ co-regulators

Many interrelationships between ER and PR pathways

- Progestins modulate estrogen action BUT also act independently
- Progestin effects on estrogen action vary in different tissues.

III. Drugs acting on Estrogen and Progesterone Receptors

A. SERMs & Antagonists SERDs

- Selective Estrogen Receptor Modulator
- Selective Estrogen Receptor Down-regulator
- Mixed estrogen receptor agonist and antagonist, effect depends on tissue type

DRUGS THAT BLOCK ER ACTIVATION AND FUNCTION ARE ANTIESTROGENS

- SERMs: Tamoxifen, Raloxifene
- SERDs: Fulvestrant

SERMs & Antagonists SERDs

1. Current hypothesis to explain the antiestrogen effects in different tissues:
    a. protein conformation
    b. co-activators for ER
    c. alternate response elements

2. Tamoxifen (a SERM)

- Tamoxifen (Nolvadex) (non-steroidal…)

MOA: Competitive partial agonist inhibitor of estradiol:

- Estrogen antagonist: effects centrally and on breast tissue
- Estrogen agonist: effects on uterus, (bone), and coagulation profile
- Therapeutic Uses: metastatic breast cancer, prophylaxis
- Adverse Effects: hot flashes, vaginal bleeding, endometrial hyperplasia
- Contraindications: Hx DVT, PE, coagulopathy
- Pharmacogenetics…..(CYP2D6)

3. Raloxifene (a SERM)

- Raloxifene (Evista)
  - Estrogen antagonist: effects centrally and on uterus and breast tissue
  - Estrogen agonist: effects on bone, lipids and coagulation profile
  - Indications: postmenopausal osteoporosis, breast cancer prophylaxis
  - Contraindications: Hx DVT, PE, coagulopathy
  - Increase hot flashes
  - Warnings: NOT for pre-menopause; incr. incidence DVT/PE

4. Fulvestrant (a SERD)

- Fulvestrant (Faslodex)
  - Estrogen antagonist: effects on breast; competes with estradiol
  - Downregulates estrogen receptor protein in breast cancer
  - Indications: metastatic breast cancer

B. Selective PR modulators (SPRMs) & Antagonists
Partial agonist/ partial antagonist

1. Ulipristal (Ella)

   SPRM

   MOA: Partial agonist/ partial antagonist at PR

   Binds to PR; prevents ovulation.

   Therapeutic use: Emergency contraception

   Effects depend on timing of the dose during the menstrual cycle (will not terminate an existing pregnancy)

B. Selective PR modulators (SPRMs) & Antagonists

2. Mifepristone (Mifeprex, RU-486)

MOA: Competitive PR antagonist

Binds to PR and inhibits activity of progesterone. (Absence of effects of progesterone leads to abortion) Also acts antagonist at cortisol receptors.

Therapeutic uses: Medical abortion; medical management of miscarriages; Cushing's syndrome

Pharm to Watch for…..

More….

- SERMS
- Selective PR modulators (SPRMs) and Antagonists

IV. Clinical Uses of Estrogens, Progesterones, and their Antagonists

Estrogens as Therapeutic Agents

1. Can be naturally derived or synthetic

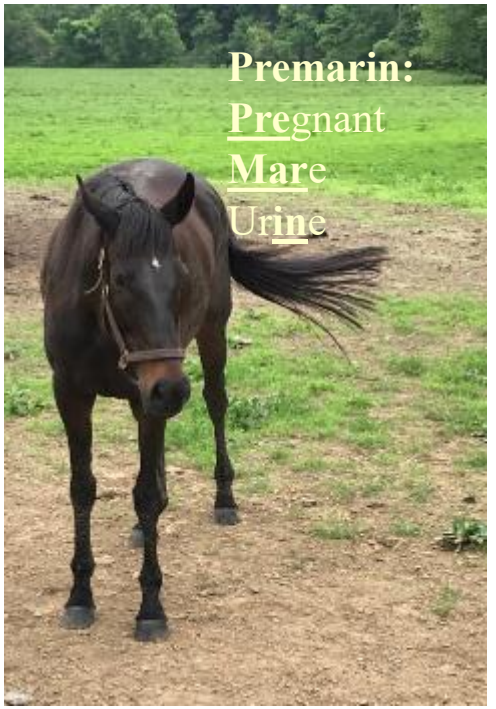
   Example: Premarin^{R}

   Naturally occurring; difficult to replicate

2. Most often oral

   Also: vaginal, transdermal

3. Used for contraception, menstrual cycle control, ovulation induction, hormone therapy



NOTE: Hormone Therapy (HT) was previously referred to as Hormone Replacement Therapy (HRT)

Progestogens as Therapeutic Agents

1. Prophylaxis against endometrial hyperplasia or carcinoma in estrogen-treated patients

2. In some cases, appropriate for solo use:
   - Induction of withdrawal bleeding in amenorrhea
   - Contraception
   - Inhibition of gonadotropin secretion– precocious puberty or treatment of endometriosis

Clinical Uses of Exogenous Hormones

A. Oral contraception

B. Other routes of administration

C. Emergency Contraception

D. Medical Abortion

This content is covered in

PHARM: Contraception & Abortion Pharmacology

E. (Postmenopausal) Hormone Therapy

E. Postmenopausal Hormone Therapy (HT)

Women's Health Initiative (WHI) was a large 5-year randomized placebo-controlled study evaluating postmenopausal HT (2002).

- Overall cohort/ preliminary analysis: increased risks of cardiovascular events
- Secondary analyses: risks influenced by age and time since menopause

- For many years, clinicians were cautious about prescribing HT

E. Postmenopausal Hormone Therapy (HT)

Current practice:

- Use for vasomotor symptoms in patient <10 years from menopause is generally safe.

- NOT used to improve CV health or to treat osteoporosis. Risk-benefit should always be considered and discussed with the patient.
- Low-dose topical (vulvar, vaginal) can be safe even with history of hormone sensitive cancers.

E. Hormone Therapy

1. FDA –approved **indications** for HT
   a. treatment of moderate to severe vasomotor symptoms
   b. treatment of moderate to severe atrophic vaginitis
   c. prevention of postmenopausal osteoporosis, NOT treatment

2. Hormone Therapy-- Estrogens

- Natural estrogens include estrone, estradiol, and conjugated equine estrogens.
- Synthetic estrogens
- Estrogens are most often administered by mouth, but they can also be administered effectively by vaginal and transdermal routes (other forms available).

3. Hormone therapy-- Progestins

Rationale for Progestin Use in HT

- Prophylaxis against endometrial hyperplasia or carcinoma in estrogen treated patients
  - Adding progestin attenuates/lessens endometrial cancer risk
- Continuous combined, continuous sequential– both are used.

Hormone Therapy

4. Guiding Principles for ET/EPT therapy
   a. Lowest effective dose
   b. Shortest duration required
   c. Type/route/regimen of E and P
   d Individualized risks and benefits
   e Patient satisfaction with therapy

Hormone Therapy

5. **Contraindications** to ET/EPT
   a. Hormone-sensitive cancer (exceptions)
   b. Unexplained uterine bleeding
   c. Active liver disease (esp. with oral ET)
   d. History of DVT/PE
   e. Confirmed CVD

Hormone Therapy

6. Common **adverse effects** of ET/EPT
   a. Uterine bleeding
   b. Breast tenderness
   c. Nausea/Abdominal bloating

F. Breast cancer therapeutics

You have had a CBL case of a patient with breast cancer. (Block Pulm)

G. Osteoporosis prophylaxis and treatment

You will have a lecture on osteoporosis pharmacology (Block MSD)

Summary

- Review of estrogen and progesterone physiology in context of understanding drug action and adverse effects
- Discussion of estrogen and progesterone receptors and tissue specificity of isoforms
- Drugs acting on estrogen and progesterone receptors: SERMs, SERDs, SPRMs
- Clinical uses



Pregnant mare



Mare with foal (1 week old)



Foal at 5.5 years old; retired racer; current career as eventing horse.

That's all!